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What Is an AI Engineer? (And How to Become One)

Written by Coursera Staff • Updated on May 28, 2026
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Learn what an artificial intelligence engineer does and how you can get into this exciting career field.



Key takeaways

Artificial intelligence (AI) engineers use their knowledge of AI and machine learning to develop computer applications and systems.

- The median total salary for an AI engineer in the United States is $138,000 [1].
- AI engineering has various applications, including recommender systems on streaming platforms like Netflix, self-driving cars, surgical robots, and more.
- You can start a career in AI engineering by first learning technical skills like programming and statistics, applying for entry-level jobs, and building your work experience in the field.

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What is AI engineering?

Artificial intelligence engineers are individuals who use AI and machine learning techniques to develop applications and systems that can help organizations increase efficiency, cut costs, increase profits, and make better business decisions.

By 2030, AI could contribute up to $19.9 trillion to the global economy and result in a 3.5 percent increase in global GDP, according to an IDC report [2]. This projected growth means organizations are turning to AI to help power their business decisions and increase efficiency.

AI engineering focuses on developing the tools, systems, and processes that enable artificial intelligence to be applied in the real world. Any application where machines mimic human functions, such as solving problems and learning, can be considered artificial intelligence. Algorithms are “trained” by data, which helps them learn and perform better.

Everyday examples of AI engineering

You may have encountered the results of AI engineering when you use Netflix, Spotify, or YouTube, where machine learning customizes suggestions based on your behavior. Another popular example is in transportation, where self-driving cars are driven by AI and machine learning technology. It's especially useful in the health care industry because AI can power robots to perform surgery and generate automated image diagnoses.

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AI engineer responsibilities: What do AI engineers do officially?

AI engineers play an important role in organizations that use AI. They chart the AI strategy and define the problems to be solved with AI. They're in charge of building AI development and production infrastructure and then implementing it. Here are some specific tasks and responsibilities of an AI engineer:

- Create and manage the AI development and production infrastructure
- Conduct statistical analysis and interpret the results to guide and optimize the organization's decision-making process
- Automate AI infrastructures for the data science team
- Build AI models from scratch and help product managers and other stakeholders with analysis and implementation
- Transform machine learning models into APIs that can be integrated with other applications
- Collaborate across teams to help with AI adoption and best practices

Are you a software developer looking to leverage AI tools for increased productivity and competitiveness in the job market? Consider enrolling in Microsoft's Generative AI for Software Developers Specialization.



AI engineer salary and job outlook

AI engineering is a specialized field that has promising job growth and tends to pay well.

According to Glassdoor, the median total salary for an AI engineer is $138,000 in the United States as of October 2025 [1]. This figure includes base salary and additional pay, which may represent profit-sharing, commissions, bonuses, or other compensation. Falling under the categories of computer and information research scientists, AI engineers have a median salary of $140,910, according to the US Bureau of Labor Statistics (BLS) [3].

There is a projected job growth of 20 percent between 2024 and 2034, which is much faster than the average for all occupations [3]. According to a Q1 2025 job market analysis by Veritone, AI/ML engineer jobs are one of the fastest-growing AI roles, showing a 41.8 percent year-on-year increase [4].

AI engineers typically work for tech companies like Google, IBM, and Meta, among others, helping them improve their products, software, operations, and delivery. They may also be employed more and more in government and research facilities that work to improve public services.

How to become an AI engineer

Becoming an AI engineer requires basic computer, information technology (IT), and math skills, as these are critical to maneuvering artificial intelligence programs. Here's how you can get started in AI engineering.

1. Learn the skills needed.

You'll need to build your technical skills, including knowledge of the tools that AI engineers typically use.

- **Programming:** You'll want to learn programming languages such as Python, R, Java, and C++ so you can build and implement models.
- **Probability, statistics, and linear algebra:** These are needed to implement different AI and machine learning models.
- **Big data technologies:** AI engineers work with large amounts of data, so you'll be required to know Apache Spark, Hadoop, and MongoDB to manage it all.
- **Algorithms and frameworks:** You'll want to understand machine learning algorithms such as linear regression and Naive Bayes, as well as deep learning algorithms, such as recurrent neural networks and generative adversarial networks, and be able to implement them with a framework. Common AI frameworks include Theano, TensorFlow, Caffe, Keras, and PyTorch.

You can learn these skills through online courses or boot camps specially designed to help you launch your career in artificial intelligence.

Not sure where to start? Consider enrolling in the University of Michigan's Python for Everybody Specialization to learn how to program and analyze data with Python in just two months. For the basics of machine learning, meanwhile, explore Stanford and DeepLearning.AI's Machine Learning Specialization.



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Do I need a degree to become an AI engineer?

No, you don’t necessarily need a degree to become an AI engineer. Increasingly, people are using Professional Certificate programs to learn the skills they need and prepare for interviews.

Still, many companies require at least a bachelor’s degree for entry-level jobs. Common majors include computer science and information technology. Jobs in AI are competitive, but if you can demonstrate you have a strong set of the right skills and interview well, then you can launch your career as an AI engineer.



2. Apply for jobs.

Applying for a job can be intimidating when you have little to no experience in a field. But it might be helpful to know that people get hired every day for jobs with no experience. There are many ways to enhance your resume for success in the job search. For AI engineering jobs, you'll want to highlight specific projects you've worked on for jobs or classes that demonstrate your broad understanding of AI engineering.

Once you've applied for jobs, you'll hopefully get an interview. Don't be discouraged if you apply for dozens of jobs and don't hear back—data science, in general, is such an in-demand (and lucrative) career field that companies can receive hundreds of applications for one job.

Preparing for the interview requires practice and preparation, especially for tech jobs like AI engineer. You'll want to brush up on your interview skills, so you can prove to hiring managers that you're perfect for the job. You'll be expected to explain your reasoning for developing, deploying, and scaling specific algorithms. These interviews can get very technical, so be sure you can clearly explain how you solved a problem and why you chose to solve it that way.

Read more: Job Search Guide: Resources for Your Next Career Move

3. Get work experience.

There may be several rounds of interviews, even for an entry-level position or internship. But if you land a job, then it's time to prove yourself and learn as much as possible. You'll be able to apply the skills you learned toward delivering business insights and solutions that can change people's lives, whether it is in health care, entertainment, transportation, or consumer product manufacturing.

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Article sources

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